

LISTA for Sparse Channel Estimation: Key Findings

NMSE Performance

LISTA: -25 dB (saturates)

OMP: -37 dB | FISTA: -34 dB

⇒ **LISTA trails by 12--27 dB**

BER Performance (ZF, 16-QAM)

LISTA matches or beats OMP

at $\text{SNR} \geq 15$ dB ($p < 0.05$)

⇒ **BER advantage despite worse NMSE**

Mechanism: Error Concentration

LISTA concentrates 100.0% of estimation error on true tap locations
(vs. 95.2% OMP, 92.4% ISTA) → $267\times$ less non-support error

Learned threshold schedule is dominant (+14--18 dB); hard thresholding outperforms soft by 7.1 dB

Use LISTA for throughput-critical ZF systems; OMP/FISTA for NMSE-critical tasks